Project Documentation

Project Title:LibSpace-A Robust Backend for Library Operations

1. **Introduction**:
   - **Project Title:**LibSpace-A Robust Backend for Library Operations
   - **Team ID:**SWTID-2026-6809
   - **Team Leader:**Navmitha afrin S (navmithaafrin@gmail.com)
   - **Team Members:**
     1)Sahana Parveen H (parveensahana821@gmail.com)
     2)Jayavarthini T (jayavarthinitheivendran@gmail.com)
     3)Palani praba D (palanipraba15@gmail.com)
     4)Abinaya R (abinaya8648@gmail.com)
2. **Project Overview:**

Purpose:

The purpose of Libspace is to develop a backend system that manages library book records and user accounts

securely.It demonstrates user authentication,secure API development,database integration,and CRUD operations.

Features:

- User registration and login
- Secure password hashing using bcrypt
- JWT-based authentication
- Add,update,and delete books
- View book list and get book by ID • One user can manage multiple books

3. Architecture:

API Layer(Routes)

Handles incoming HTTP requests and API endpoints using Node.js and Express.js.

- /api/auth/register
- /api/auth/login
- /api/book/add
- /api/book/:id

Business Logic Layer(Controllers)

- Controllers implement core application logic including authentication,password hashing,JWT generation,and CRUD operations.

Data Access Layer(Database/Models)

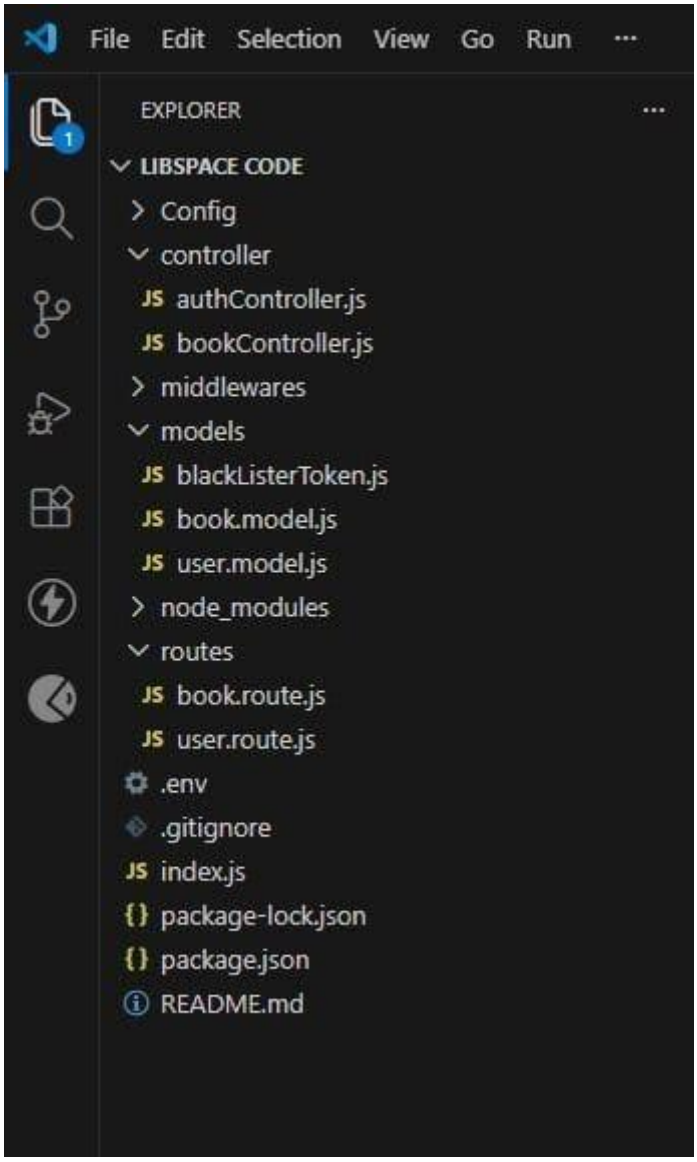
- The data layer manages database interaction using Mongodb and Mongoose.

- Models represent Users and Books collections.

4. Technologies Used:

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT(JSON Web Token)
- bcrypt
- Thunder client • Visual Studio Code **5. Installation:**
- Install Dependencies **npm install**
- Environment Variables(.env)
  PORT=5000
  MONGO_URI=mongodb;//127.0.0.1;27017/libs pace
  JWT_SECRET=your_secret_key
- Run the Server **npm start or**
  npm run dev

6. Folder Structure:



7. API Documentation:

Users APIs

- POST/api/auth/register
- POST/api/auth/login
- DELETE/api/auth/delete
- POST/api/auth/logout

Book APIs

- POST/api/book/add
- GET/api/book
- GET/api/book/:id

- PUT/api/book/:id
- DELETE/api/book/:id

8. Authentication:

LibSpace uses JWT for authentication.After login,the server generates a token which must be sent in request headers for protected routes.

Authorization: Bearer <token>

9. Database Design

Users Collection Fields

- name
- email
- password
- createdAt
- updatedAt

Books Collection Fields

- title
- author
- url
- user(Objected reference)

10.Testing:

1. Tested using Thunder client
2. Verified using registration and login
3. Verified book CRUD operations
4. Verified token authentication

11.Screenshots:

Fig 1.1 Successful Connection to MongoDB

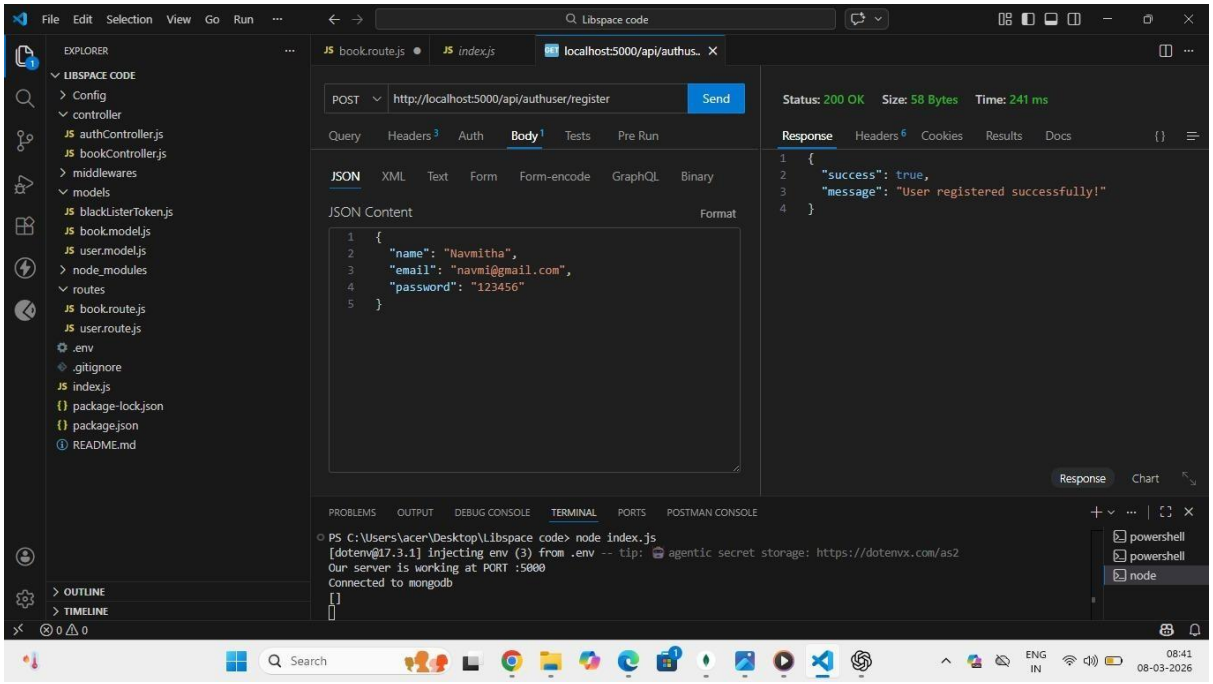


Fig 1.2 Displaying Book Title and Author

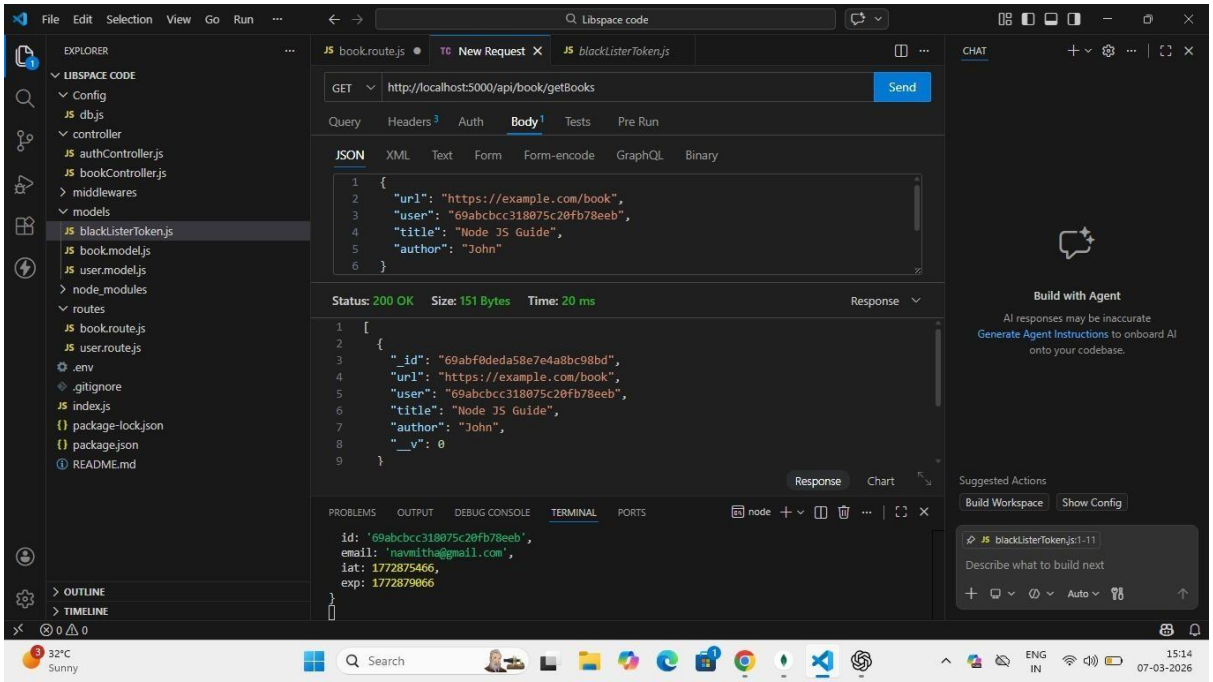


Fig 1.3 Adding a New Book

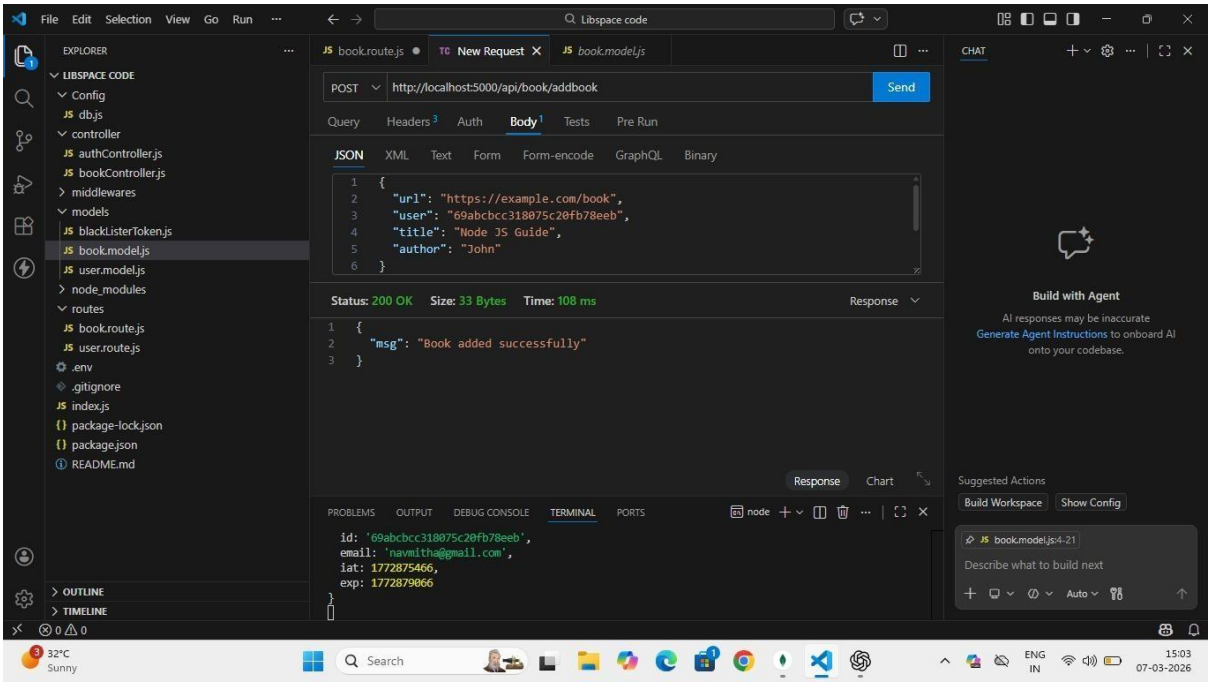


Fig 1.4 Deleting a Book

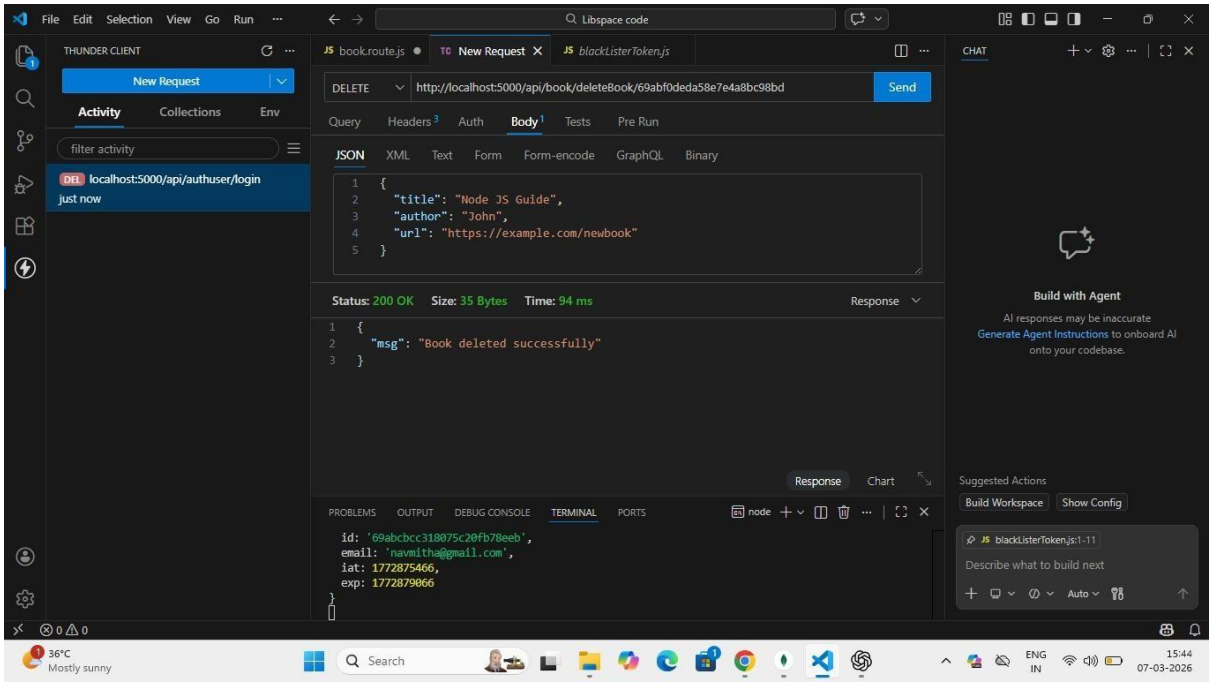
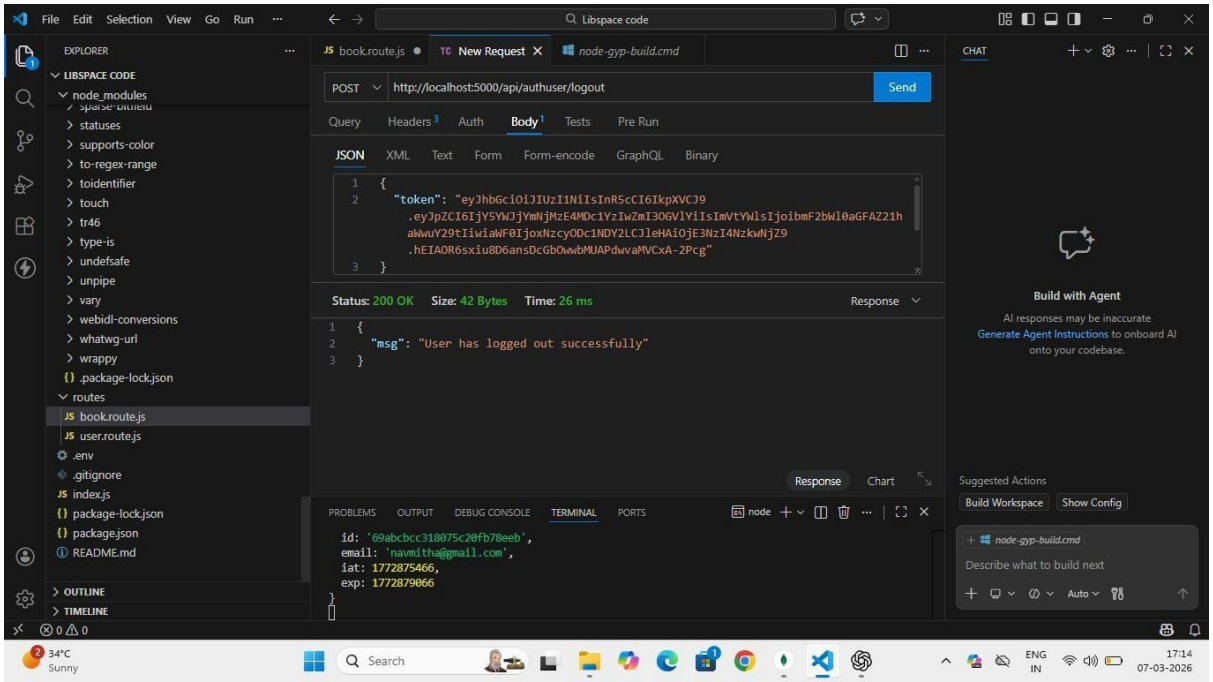


Fig 1.5 User Logout Response



12.Known Issues:

- No frontend interface
- Requires Thunder client for testing • Token blacklist optimization needed

13.Future Enhancements:

- Frontend integration with React
- Role-based authentication
- Cloud database deployment
- Book file upload support • Mobile application support

14.Demo Video Link: https://drive.google.com/file/d/1hi-XaFOBFLYv056XNjzJLxri76PH09MF/view?usp=sharing

15.Libspace Project Code:

https://drive.google.com/drive/folders/1_bYY6CaN1uWB6pdFYpwfyJqSS0CgI-MO